# SPRINT 3

## Gestió de taules, índex i vistes

Erik Subirats Cruz

# NIVELL 1 ⭐

## Exercici 1

La teva tasca és dissenyar i crear una taula anomenada "credit_card" que emmagatzemi detalls crucials sobre les targetes de crèdit. La nova taula ha de ser capaç d'identificar de manera única cada targeta i establir una relació adequada amb les altres dues taules ("transaction" i "company"). Després de crear la taula serà necessari que ingressis la informació del document denominat "dades_introduir_credit". Recorda mostrar el diagrama i realitzar una breu descripció d'aquest.

En primer lloc, he creat la taula *credit_card*, definint la columna *id* com a clau primària (PRIMARY KEY), ja que identifica de manera única cada targeta. A més, he afegit la resta de columnes, totes elles com a VARCHAR:

```
CREATE TABLE  IF NOT EXISTS credit_card (
    id VARCHAR(15) PRIMARY KEY,
    iban VARCHAR(50),
    pan VARCHAR(20),
    pin VARCHAR(4),
    cw VARCHAR(3),
    expiring_date VARCHAR(8)
);
```

Output

Action Output

| # | Time | Action | Message |
|---|---|---|---|
| 1 | 12:56:23 | USE transactions | 0 row(s) affected |
| 2 | 12:56:24 | CREATE TABLE  IF NOT EXISTS credit_card (  id VARCHAR(15) PRIMARY KEY UNIQUE,  iban VARCHAR(50),  pan VARCHAR(20),  pin VARC... | 0 row(s) affected |

A continuació, he executat el fitxer *datos_introducir_sprint3_credit*, que conté les instruccions INSERT INTO, per tal d'introduir totes les dades dins la nova taula *credit_card*:

```
4844 • INSERT INTO credit_card (id, iban, pan, pin, cw, expiring_date) VALUES ('CcS-9409', 'XX7185807133413424309528 43', '8860296285646892', '3803', '800', '06/27/26');
4845 • INSERT INTO credit_card (id, iban, pan, pin, cw, expiring_date) VALUES ('CcS-9410', 'XX8627374564650022040900 5', '8119860175041790', '1632', '300', '11/25/28');
4846 • INSERT INTO credit_card (id, iban, pan, pin, cw, expiring_date) VALUES ('CcS-9411', 'XX5842395206260291030813 6', '5760977857571518', '2288', '579', '05/29/26');
4847 • INSERT INTO credit_card (id, iban, pan, pin, cw, expiring_date) VALUES ('CcS-9412', 'XX1088506265657336190379 63', '4327349292666841', '3086', '497', '07/27/25');
4848 • INSERT INTO credit_card (id, iban, pan, pin, cw, expiring_date) VALUES ('CcS-9413', 'XX9611186991744412228825894', '7082042823580232', '1622', '262', '06/28/26');
4849 • INSERT INTO credit_card (id, iban, pan, pin, cw, expiring_date) VALUES ('CcS-9414', 'XX6857049036447665124450 77', '1630313635211518', '8005', '182', '06/26/28');
4850 • INSERT INTO credit_card (id, iban, pan, pin, cw, expiring_date) VALUES ('CcS-9415', 'XX1882435345324497873690 23', '1613956819067195', '3909', '372', '06/25/27');
4851 • INSERT INTO credit_card (id, iban, pan, pin, cw, expiring_date) VALUES ('CcS-9416', 'XX3752792408355906821460 87', '9224649624558191', '2998', '931', '08/29/29');
4852 • INSERT INTO credit_card (id, iban, pan, pin, cw, expiring_date) VALUES ('CcS-9417', 'XX2792622668768413111467 1', '6780765170929289', '7627', '915', '12/27/28');
4853 • INSERT INTO credit_card (id, iban, pan, pin, cw, expiring_date) VALUES ('CcS-9418', 'XX2739678141160226482766 43', '2288072132045684', '5442', '902', '08/26/25');
4854 • INSERT INTO credit_card (id, iban, pan, pin, cw, expiring_date) VALUES ('CcS-9419', 'XX2250734081992150529248 15', '1399718767204335', '1351', '126', '07/28/28');
4855 • INSERT INTO credit_card (id, iban, pan, pin, cw, expiring_date) VALUES ('CcS-9420', 'XX1913946193207515762255 27', '3041468287205784', '8121', '655', '01/25/25');
4856 • INSERT INTO credit_card (id, iban, pan, pin, cw, expiring_date) VALUES ('CcS-9421', 'XX2780968844447284273922 52', '1695330438148463', '9515', '699', '07/29/28');
4857 • INSERT INTO credit_card (id, iban, pan, pin, cw, expiring_date) VALUES ('CcS-9422', 'XX5196206786313052409049 63', '8005448887380420', '2920', '379', '06/28/26');
4858 • INSERT INTO credit_card (id, iban, pan, pin, cw, expiring_date) VALUES ('CcS-9423', 'XX4874936424323730779972 041', '3335566052980562', '7284', '958', '11/25/28');
4859 • INSERT INTO credit_card (id, iban, pan, pin, cw, expiring_date) VALUES ('CcS-9424', 'XX9352994124781334200823 64', '4113724797997789', '6873', '862', '10/29/29');
```

Output

Action Output

| # | Time | Action | Message |
|---|---|---|---|
| 4985 | 13:18:48 | INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS-9566', 'XX3270463329268112780556 30', '6161849983263025', '6889',... | 1 row(s) affected |
| 4986 | 13:18:48 | INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS-9567', 'XX8563817838725386032800 64', '0745947204130257', '4955',... | 1 row(s) affected |
| 4987 | 13:18:48 | INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS-9568', 'XX3865365478223814317476 59', '2296680186377148', '8326',... | 1 row(s) affected |
| 4988 | 13:18:48 | INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS-9569', 'XX3052727289640478973615 63', '0734570924276022', '6083',... | 1 row(s) affected |
| 4989 | 13:18:48 | INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS-9570', 'XX6339420102582861815165 73', '7233675965634598', '5619',... | 1 row(s) affected |
| 4990 | 13:18:48 | INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS-9571', 'XX4849153824371419965794 68', '1265066247957405', '6585',... | 1 row(s) affected |
| 4991 | 13:18:48 | INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS-9572', 'XX4054481862697303884045 08', '8296622848225053', '1995',... | 1 row(s) affected |
| 4992 | 13:18:48 | INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS-9573', 'XX9653389310530889019249 06', '9099040179637171', '2277',... | 1 row(s) affected |
| 4993 | 13:18:48 | INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS-9574', 'XX6627617583614326865207 75', '3171178331318656', '5084',... | 1 row(s) affected |
| 4994 | 13:18:48 | INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS-9575', 'XX4998812160736265711964 73', '8086033007377786', '7256',... | 1 row(s) affected |
| 4995 | 13:18:48 | INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS-9576', 'XX5297109305877126417200 7', '4554226978064107', '4884', '... | 1 row(s) affected |
| 4996 | 13:18:48 | INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS-9577', 'XX1589140785948086331470 86', '8110437271856107', '5864',... | 1 row(s) affected |

Finalment, he modificat la taula *transaction* per establir una relació entre les dues taules. Concretament, la columna *credit_card_id* de la taula *transaction* s'ha definit com a clau forana (FOREIGN KEY) que fa referència a la columna *id* de la taula *credit_card*.

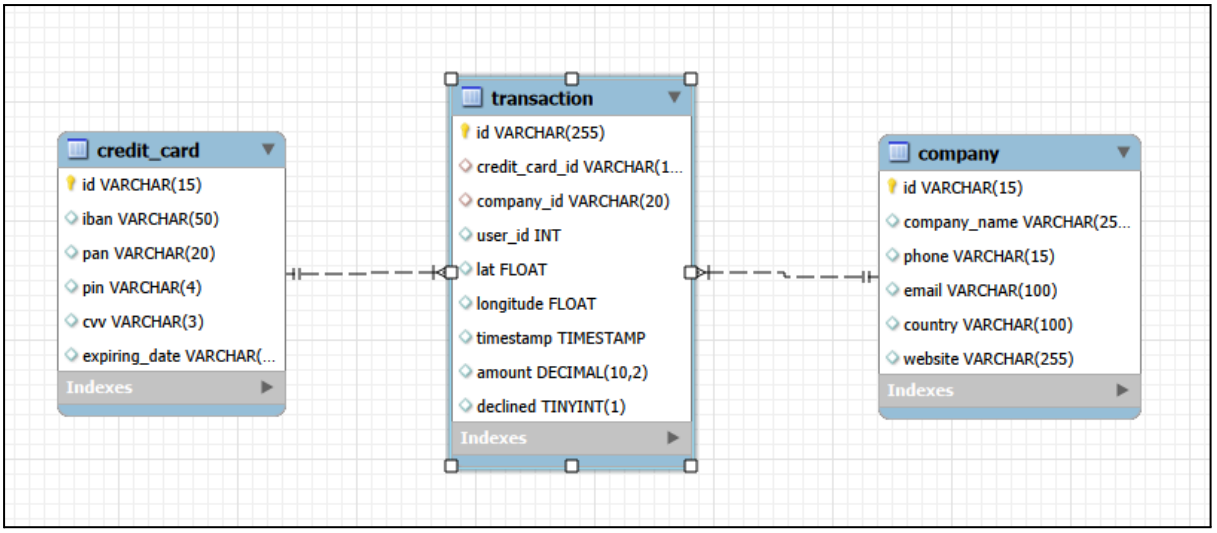
```
5019 • ALTER TABLE transaction
5020   ADD FOREIGN KEY (credit_card_id) REFERENCES credit_card(id);
5021
```

Output

Action Output

| # | Time | Action | Message |
|---|---|---|---|
| 4999 | 13:18:48 | INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcS-9580', 'XX7812588898519508066773 58', '5541182364498931', '9273',... | 1 row(s) affected |
| 5000 | 13:20:45 | ALTER TABLE transaction ADD FOREIGN KEY (credit_card_id) REFERENCES credit_card(id) | 100000 row(s) affected Records: 100000 Duplicates: 0 Warnings: 0 |

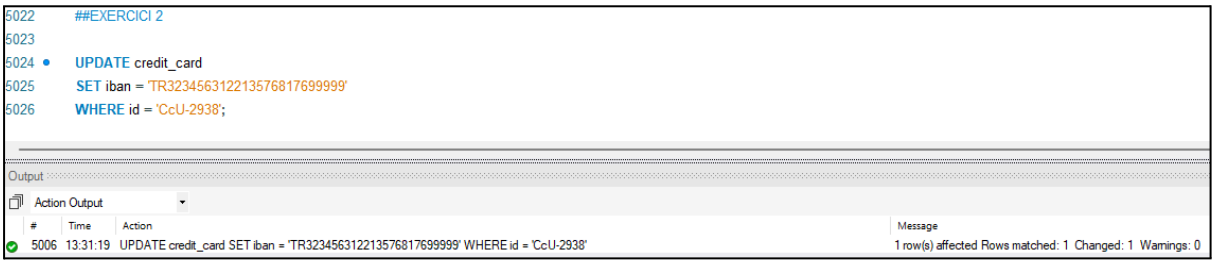
Pel que fa al diagrama, mostra clarament l'estructura de la base de dades i les relacions entre les taules. Podem veure com la relació entre les taules *credit_card* i *transactions* és de tipus 1:N, ja que una targeta de crèdit pot estar associada a múltiples transaccions, mentre que cada transacció només pot estar associada a una única targeta.




# Exercici 2

El departament de Recursos Humans ha identificat un error en el número de compte associat a la targeta de crèdit amb ID CcU-2938. La informació que ha de mostrar-se per a aquest registre és: TR323456312213576817699999. Recorda mostrar que el canvi es va realitzar.

Per tal de resoldre l'error fem servir la instrucció UPDATE i SET:



```
##EXERCICI 2

UPDATE credit_card
SET iban = 'TR323456312213576817699999'
WHERE id = 'CcU-2938';
```

Output

Action Output

| # | Time | Action | Message |
|---|---|---|---|
| 5006 | 13:31:19 | UPDATE credit_card SET iban = 'TR323456312213576817699999' WHERE id = 'CcU-2938' | 1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0 |

Comprovació utilitzant SELECT:

```
5030 •  SELECT id, iban
5031    FROM credit_card
5032    WHERE id = 'CcU-2938';
5033
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

| id | iban |
|---|---|
| CcU-2938 | TR3234563122135768176999999 |
| NULL | NULL |

credit_card 11

Output

Action Output

| # | Time | Action | Message |
|---|---|---|---|
| 5008 | 13:32:53 | SELECT id, iban FROM credit_card WHERE id = 'CcU-2938' | 1 row(s) returned |

# Exercici 3

En la taula "transaction" ingressa una nova transacció amb la següent informació:

| | |
|---|---|
| Id | 108B1D1D-5B23-A76C-55EF-C568E49A99DD |
| credit_card_id | CcU-9999 |
| company_id | b-9999 |
| user_id | 9999 |
| lat | 829.999 |
| longitude | -117.999 |
| amount | 111.11 |
| declined | 0 |

He ingressat les dades que em demana, però m'ha donat ERROR (*Error Code: 1452. Cannot add or update a child row: a foreign key constraint fails (transactions.transaction, CONSTRAINT transaction_ibfk_1 FOREIGN KEY (company_id) REFERENCES company (id))*):

```
5038 •  INSERT INTO transaction (id, credit_card_id, company_id, user_id, lat, longitude,  amount, declined) VALUES ('108B1D1D-5B23-A76C-55EF-C568E49A99DD', 'CcU-9999', 'b-9999', '9999', '829.999', '-117.999', '111.11', '0');
5039
```

Output

Action Output

| # | Time | Action | Message |
|---|---|---|---|
| 5013 | 00:27:00 | INSERT INTO transaction (id, credit_card_id, company_id, user_id, lat, longitude,  amount, declined) VALUES ('108B1D1D-5B23-A76C-55EF-C568E49... | Error Code: 1452. Cannot add or update a child row: a foreign key constraint fails (`transactions`.`transaction` |

Aquest error indica que els valors de *company_id* i *credit_card_id* no existien a les seves respectives taules (*company* i *credit_card*). Comprovem-ho:

```
5042 • SELECT *
5043   FROM credit_card
5044   WHERE id = 'CcU-9999';
5045
5046 • SELECT *
5047   FROM company
5048   WHERE id = 'b-9999';
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

| id | company_name | phone | email | country | website |
|---|---|---|---|---|---|
| NULL | NULL | NULL | NULL | NULL | NULL |

credit_card 17 | company 18

Output

Action Output

| # | Time | Action | Message |
|---|---|---|---|
| 5018 | 00:32:35 | SELECT * FROM company WHERE id = 'b-9999' | 0 row(s) returned |
| 5019 | 00:32:37 | SELECT * FROM credit_card WHERE id = 'CcU-9999' | 0 row(s) returned |

Afegeixo, doncs, a les seves respectives taules, els valors de *id* tant de la *credit_card* com de la *company*:

```
5052 • INSERT INTO credit_card (id, iban, pan, pin, cvv, expiring_date) VALUES ('CcU-9999', NULL, NULL, NULL, NULL, NULL);
5053 • INSERT INTO company (id, company_name, phone, email, country, website) VALUES ('b-9999', NULL, NULL, NULL, NULL, NULL);
5054
5055
```

Output

Action Output

| # | Time | Action | Message |
|---|---|---|---|
| 5021 | 00:37:54 | INSERT INTO credit_card ( id, iban, pan, pin, cvv, expiring_date ) VALUES ( 'CcU-9999'... | 1 row(s) affected |
| 5022 | 00:40:32 | INSERT INTO company (id, company_name, phone, email, country, website) VALUES ('b-9999', NULL, N... | 1 row(s) affected |

Un cop existents els registres a les taules *company* i *credit_card*, ja podem ingressar la transacció:

```
5056 • INSERT INTO transaction (id, credit_card_id, company_id, user_id, lat, longitude, amount, declined) VALUES ('108B1D1D-5B23-A76C-55EF-C568E49A99DD', 'CcU-9999', 'b-9999', '9999', '829.999', '-117.999', '111.11', '0');
5057
```

Output

Action Output

| # | Time | Action | Message |
|---|---|---|---|
| 5022 | 00:40:32 | INSERT INTO company (id, company_name, phone, email, country, website) VALUES ('b-9999', NULL, NULL, NULL, NULL, NULL) | 1 row(s) affected |
| 5023 | 00:43:14 | INSERT INTO transaction (id, credit_card_id, company_id, user_id, lat, longitude, amount, declined) VALUES ('108B1D1D-5B23-A76C-55EF-C568E49A99DD', 'CcU-9... | 1 row(s) affected |

# Exercici 4

Des de recursos humans et sol·liciten eliminar la columna "pan" de la taula credit_card. Recorda mostrar el canvi realitzat.

Elimino la columna pan amb la instrucció DROP COLUMN:

```
# Elimino la columna pan de credit_card
ALTER TABLE credit_card
DROP COLUMN pan;
```

Output

Action Output

| # | Time | Action | Message |
|---|---|---|---|
| 5023 | 00:43:14 | INSERT INTO transaction (id, credit_card_id, company_id, user_id, lat, longitude, amount, declined) VALUES ('108B1D1D-5B23-A76C-55EF-C568E49A99DD', 'CcU-9... | 1 row(s) affected |
| 5024 | 00:54:53 | ALTER TABLE credit_card DROP COLUMN pan | 0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0 |

Comprovació amb DESCRIBE:

```
DESCRIBE credit_card;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| Field | Type | Null | Key | Default | Extra |
|---|---|---|---|---|---|
| id | varchar(15) | NO | PRI | NULL | |
| iban | varchar(50) | YES | | NULL | |
| pin | varchar(4) | YES | | NULL | |
| cvv | varchar(3) | YES | | NULL | |
| expiring_date | varchar(8) | YES | | NULL | |

Result 19

Output

Action Output

| # | Time | Action | Message |
|---|---|---|---|
| 5024 | 00:54:53 | ALTER TABLE credit_card DROP COLUMN pan | 0 row(s) affected R |
| 5025 | 00:57:02 | DESCRIBE credit_card | 5 row(s) returned |

# NIVELL 2 ⭐⭐

## Exercici 1

Elimina de la taula transaction el registre amb ID 000447FE-B650-4DCF-85DE-C7ED0EE1CAAD de la base de dades.

Elimino el registre amb la instrucció DELETE:

```
5071 •  DELETE FROM transaction
5072    WHERE id='000447FE-B650-4DCF-85DE-C7ED0EE1CAAD';
```

Output

Action Output

| # | Time | Action | Message |
|---|---|---|---|
| 5025 | 00:57:02 | DESCRIBE credit_card | 5 row(s) returned |
| 5026 | 01:05:43 | DELETE FROM transaction WHERE id='000447FE-B650-4DCF-85DE-C7ED0EE1CAAD' | 1 row(s) affected |

Comprovació de l'eliminació:

```
5074    #Comprovació
5075 •  SELECT *
5076    FROM transaction
5077    WHERE id = '000447FE-B650-4DCF-85DE-C7ED0EE1CAAD';
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

| id | credit_card_id | company_id | user_id | lat | longitude | timestamp | amount | declined |
|---|---|---|---|---|---|---|---|---|
| NULL | NULL | NULL | NULL | NULL | NULL | NULL | NULL | NULL |

transaction 20

Output

Action Output

| # | Time | Action | Message |
|---|---|---|---|
| 5026 | 01:05:43 | DELETE FROM transaction WHERE id='000447FE-B650-4DCF-85DE-C7ED0EE1CAAD' | 1 row(s) affected |
| 5027 | 01:07:54 | SELECT * FROM transaction WHERE id = '000447FE-B650-4DCF-85DE-C7ED0EE1CAAD' | 0 row(s) returned |

## Exercici 2

La secció de màrqueting desitja tenir accés a informació específica per a realitzar anàlisi i estratègies efectives. S'ha sol·licitat crear una vista que proporcioni detalls clau sobre les companyies i les seves transaccions. Serà necessària que creïs una vista anomenada VistaMarketing que contingui la següent informació: Nom de la companyia. Telèfon de contacte. País de residència. Mitjana de compra realitzat per cada companyia. Presenta la vista creada, ordenant les dades de major a menor mitjana de compra.

Creació de la vista *VistaMarketing* amb la instrucció CREATE VIEW:

```sql
CREATE VIEW VistaMarketing AS
SELECT
    c.company_name,
    c.phone,
    c.country,
    ROUND(AVG(t.amount), 2) AS purchase_avg
FROM company c
JOIN transaction t ON c.id = t.company_id
WHERE t.declined = 0
GROUP BY c.id, c.company_name;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| company_name | phone | country | purchase_avg |
|---|---|---|---|
| Eget Tincidunt Dui Institute | 05 35 93 32 44 | Netherlands | 254.62 |
| Neque Tellus Imperdiet Corp. | 09 15 42 22 11 | Ireland | 263.86 |
| Fusce Corp. | 08 14 97 58 85 | United States | 265.12 |
| Mus Aenean Eget Foundation | 06 25 15 52 43 | Sweden | 256.00 |
| Aliquam Iaculis Lacus Corp. | 04 43 07 91 26 | Belgium | 260.20 |

Result 29

Output

Action Output

| # | Time | Action | Message |
|---|---|---|---|
| 5036 | 01:44:49 | CREATE VIEW VistaMarketing AS SELECT c.company_name, c.phone, c.country, ROUND(AVG(t.amount), 2) AS purchase_avg FROM company c JOIN tra... | 0 row(s) affected |

Presentació de la vista creada, amb l'ordre demanat:

```sql
SELECT *
FROM VistaMarketing
ORDER BY purchase_avg DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| company_name | phone | country | purchase_avg |
|---|---|---|---|
| Ac Fermentum Incorporated | 06 85 56 52 33 | Germany | 284.91 |
| Pretium Neque Corp. | 07 77 48 55 28 | Australia | 275.58 |
| Urna Convallis Associates | 06 01 24 77 04 | United States | 273.57 |
| At Associates | 09 56 61 10 65 | New Zealand | 272.74 |
| Metus Vitae Associates | 08 25 44 40 66 | Australia | 270.05 |

VistaMarketing 36

Output

Action Output

| # | Time | Action | Message |
|---|---|---|---|
| 5045 | 01:54:59 | SELECT * FROM VistaMarketing ORDER BY purchase_avg DESC | 101 row(s) returned |

# Exercici 3

Filtra la vista VistaMarketing per a mostrar només les companyies que tenen el seu país de residència en "Germany"

```
5104 •  SELECT *
5105    FROM VistaMarketing
5106    WHERE country = 'Germany';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| company_name | phone | country | purchase_avg |
|---|---|---|---|
| Auctor Mauris Corp. | 05 62 87 14 41 | Germany | 254.68 |
| Ac Fermentum Incorporated | 06 85 56 52 33 | Germany | 284.91 |
| Nunc Interdum Incorporated | 05 18 15 48 13 | Germany | 259.32 |
| Aliquam PC | 01 45 73 52 16 | Germany | 252.96 |
| Ac Industries | 09 34 65 40 60 | Germany | 255.17 |

VistaMarketing 38

Output

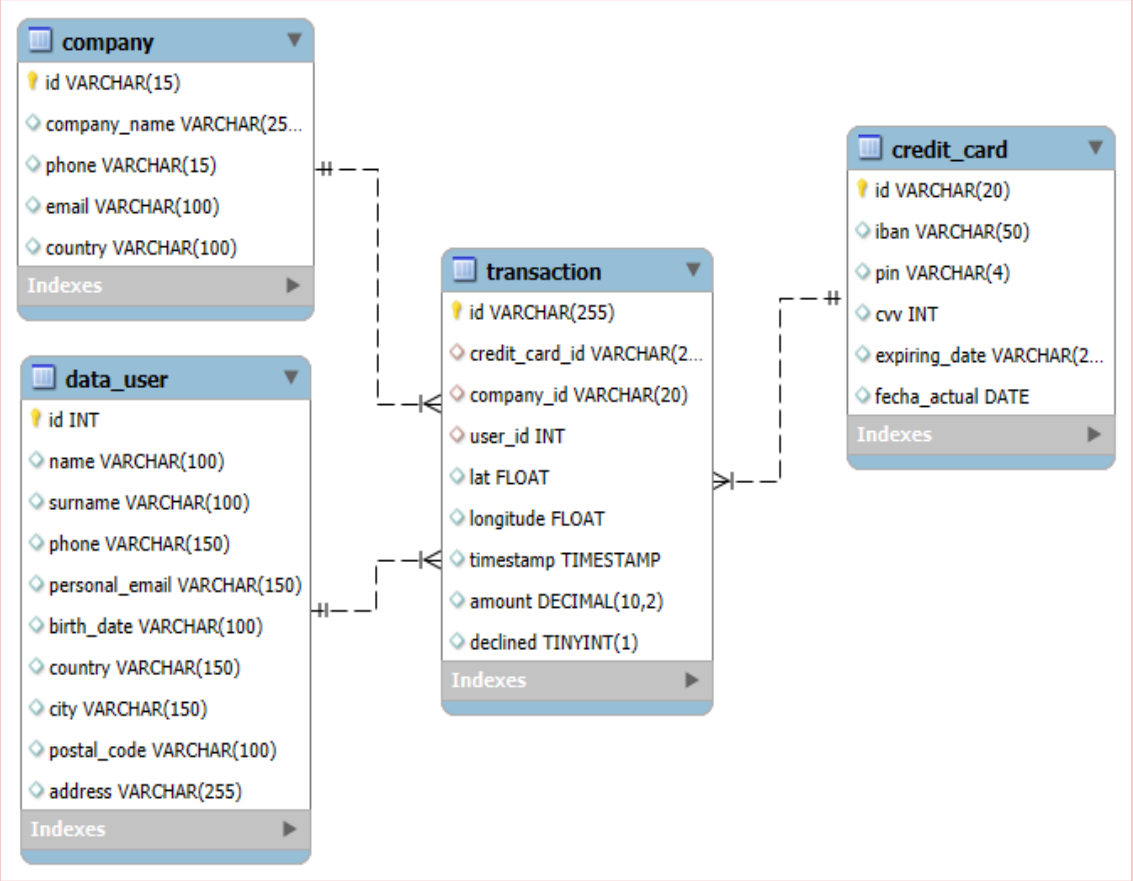
Action Output

| # | Time | Action | Message |
|---|---|---|---|
| 5047 | 01:55:49 | SELECT * FROM VistaMarketing WHERE country = 'Germany' | 8 row(s) returned |

**NIVELL 3** ⭐⭐⭐

## Exercici 1

La setmana vinent tindràs una nova reunió amb els gerents de màrqueting. Un company del teu equip va realitzar modificacions en la base de dades, però no recorda com les va realitzar. Et demana que l'ajudis a deixar els comandos executats per a obtenir el següent diagrama:




Abans de res, creo la taula *data_user* tenint en compte el document *estructura datos user.sql* i hi afegeixo les dades:

```
5112 CREATE TABLE IF NOT EXISTS user (
5113     id CHAR(10) PRIMARY KEY,
5114     name VARCHAR(100),
5115     surname VARCHAR(100),
5116     phone VARCHAR(150),
5117     email VARCHAR(150),
5118     birth_date VARCHAR(100),
5119     country VARCHAR(150),
5120     city VARCHAR(150),
5121     postal_code VARCHAR(100),
5122     address VARCHAR(255)
5123 );
```

Output

Action Output

| # | Time | Action | Message |
|---|---|---|---|
| 1 | 11:02:44 | CREATE TABLE IF NOT EXISTS user ( id CHAR(10) PRIMARY KEY, name VARCHAR(100), surname VARCHAR(100), phone VARCHAR(150), email VA... | 0 row(s) affected |

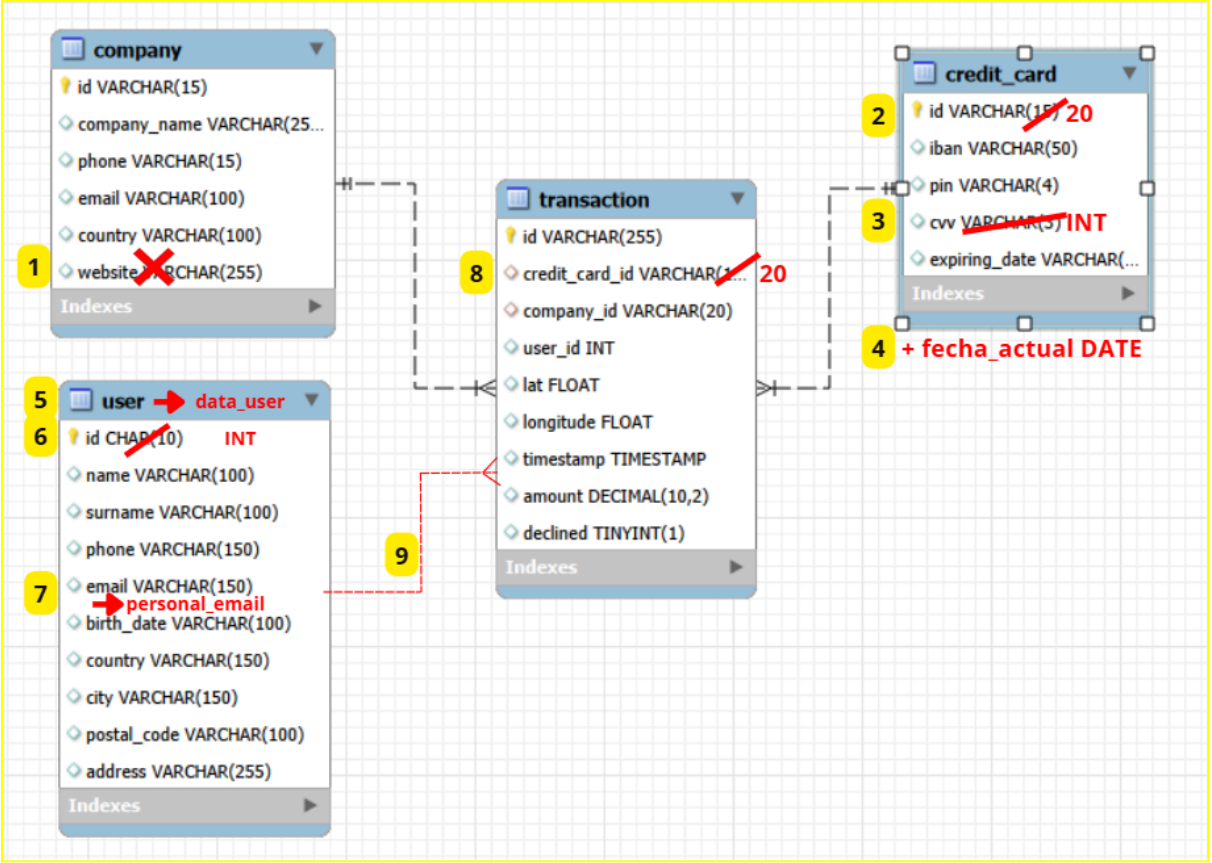
```
10113 INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES (  "4927", "Npecth", "Izfujzmh", "+60-230-443
10114 INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES (  "4928", "Paasw", "Kdxrcgta", "+75-743-607
10115 INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES (  "4931", "Vrhgue", "Npzdnfea", "+89-915-11
10116 INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES (  "4939", "Nzultp", "Lylyewdk", "+94-736-575
10117 INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES (  "4945", "Pimdet", "Zvagjirk", "+91-727-8072
10118 INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES (  "4947", "Ypyafn", "Tfiiwpnz", "+95-878-8855
10119 INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES (  "4950", "Akxgbk", "Fxtbrlub", "+96-670-322
10120 INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES (  "4952", "Fufhmw", "Sbgptspx", "+61-136-13
10121 INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES (  "4961", "Phwjfi", "Myxbvzkf", "+34-594-187
10122 INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES (  "4977", "Gjldyg", "Egqqecse", "+62-591-50
10123 INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES (  "4980", "Asgaxi", "Btardzti", "+39-155-6819
10124 INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES (  "4983", "Eremdc", "Orekppbr", "+59-545-47
10125 INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES (  "4999", "Omjnoy", "Rmuqvxgw", "+32-354-
10126
10127
```

Output

Action Output

| # | Time | Action | Message |
|---|---|---|---|
| 5000 | 11:03:59 | INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES (  "4983", "Eremdc", "Orekppbr", "+5... | 1 row(s) affected |
| 5001 | 11:03:59 | INSERT INTO user (id, name, surname, phone, email, birth_date, country, city, postal_code, address) VALUES (  "4999", "Omjnoy", "Rmuqvxgw", "... | 1 row(s) affected |

Ara és moment d'obtenir el diagrama demant fent els 9 següents canvis al meu diagrama actual:




1r canvi: eliminar columna *website* de *company*:

```
#1r canvi: eliminar columna website de company
ALTER TABLE company
DROP COLUMN website;
```

Output

Action Output

| # | Time | Action | Message |
|---|---|---|---|
| 1 | 10:06:25 | ALTER TABLE company DROP COLUMN website | 0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0 |

2n canvi: canvi del VARCHAR(15) a VARCHAR(20) a *id* de *credit_card*:

```
ALTER TABLE credit_card
MODIFY COLUMN id VARCHAR(20);
```

Output

Action Output

| # | Time | Action | Message |
|---|---|---|---|
| 5002 | 11:16:43 | ALTER TABLE company DROP COLUMN email | 0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0 |
| 5003 | 11:21:03 | ALTER TABLE credit_card MODIFY COLUMN id VARCHAR(20) | 0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0 |

3r canvi: canvi del VARCHAR a INT a *cvv* de *credit_card*:

```
#3r canvi: canvi del VARCHAR a INT a cvv de credit_card
ALTER TABLE credit_card
MODIFY COLUMN cvv INT;
```

Output

Action Output

| # | Time | Action | Message |
|---|---|---|---|
| 5003 | 11:21:03 | ALTER TABLE credit_card MODIFY COLUMN id VARCHAR(20) | 0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0 |
| 5004 | 11:21:51 | ALTER TABLE credit_card MODIFY COLUMN cvv INT | 5001 row(s) affected Records: 5001 Duplicates: 0 Warnings: 0 |

4t canvi: afegir la columna *fecha_actual* del tipus DATE a *credit_card*:

```
#4t canvi: afegir la columna fecha_actual del tipus DATE a credit_card
ALTER TABLE credit_card
ADD fecha_actual DATE;
```

Output

Action Output

| # | Time | Action | Message |
|---|---|---|---|
| 5004 | 11:21:51 | ALTER TABLE credit_card MODIFY COLUMN cvv INT | 5001 row(s) affected Records: 5001 Duplicates: 0 Warning |
| 5005 | 11:23:12 | ALTER TABLE credit_card ADD fecha_actual DATE | 0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0 |

5é canvi: canvi de nom de la taula *user* a *data_user*:

```
#5é canvi: canvi de nom de la taula user a date_user:
RENAME TABLE user TO data_user;
```

Output

Action Output

| # | Time | Action | Message |
|---|---|---|---|
| 5005 | 11:23:12 | ALTER TABLE credit_card ADD fecha_actual DATE | 0 row(s) affected Rec |
| 5006 | 11:29:45 | RENAME TABLE user TO data_user | 0 row(s) affected |

6é canvi: canvi del CHAR a INT a *id* de *data_user*:

```
#6é canvi: canvi del CHAR a INT a id de data_user:
ALTER TABLE data_user
MODIFY COLUMN id INT;
```

Output

Action Output

| # | Time | Action | Message |
|---|---|---|---|
| 5006 | 11:29:45 | RENAME TABLE user TO data_user | 0 row(s) affected |
| 5007 | 11:31:05 | ALTER TABLE data_user MODIFY COLUMN id INT | 5000 row(s) affected Records: 5000 Duplicates: 0 Warnings: 0 |

7é canvi: canvi de nom de la columna *email* a *personal_email* de *data_user*:

```
#7é canvi: canvi de nom de la columna email a personal_email de data_user:
ALTER TABLE data_user
RENAME COLUMN email TO personal_email;
```

Output

Action Output

| # | Time | Action | Message |
|---|---|---|---|
| 5007 | 11:31:05 | ALTER TABLE data_user MODIFY COLUMN id INT | 5000 row(s) affected Records: 5000 Duplicates: 0 Warni |
| 5008 | 11:33:11 | ALTER TABLE data_user RENAME COLUMN email TO personal_email | 0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0 |

8é canvi: canvi del VARCHAR(15) a VARCHAR(20) a *credit_card_id* de *transaction*:

```
#8é canvi: canvi del VARCHAR(15) a VARCHAR(20) a credit_card_id de transaction
ALTER TABLE transaction
MODIFY credit_card_id VARCHAR(20);
```

Output

Action Output

| # | Time | Action | Message |
|---|---|---|---|
| 5008 | 11:33:11 | ALTER TABLE data_user RENAME COLUMN email TO personal_email | 0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0 |
| 5009 | 11:36:26 | ALTER TABLE transaction MODIFY credit_card_id VARCHAR(20) | 0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0 |

9é canvi: establir la columna *user_id* de la taula *transaction* com a clau forana (FOREIGN KEY) que fa referència a la columna *id* de la taula *data_user*:

```
#9é canvi: establir la columna user_id de la taula transaction com a clau forana (FOREIGN KEY) que fa referència a la columna id de la taula data_user (ERROR)
ALTER TABLE transaction
ADD FOREIGN KEY (user_id) REFERENCES data_user(id);

#Busco l'error
SELECT user_id
FROM transaction
WHERE user_id NOT IN (SELECT id FROM data_user);
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| user_id |
|---|
| 9999 |

transaction 1

Output

Action Output

| # | Time | Action | Message |
|---|---|---|---|
| 5010 | 11:41:01 | ALTER TABLE transaction ADD FOREIGN KEY (user_id) REFERENCES data_user(id) | Error Code: 1452. Cannot add or update a child row: a foreign key constraint fails (`transa |
| 5011 | 11:41:54 | SELECT user_id FROM transaction WHERE user_id NOT IN (SELECT id FROM data_user) | 1 row(s) returned |

La instrucció dona ERROR perquè hi ha un valor que apareix a la taula *transaction.user_id* que no existeix a *user.id*. És per això que afegim aquest registre:

```
#Afegeixo el registre a user.id
INSERT INTO data_user (id, name, surname, phone, personal_email, birth_date, country, city, postal_code, address) VALUES ('9999', NULL, NULL, NULL, NULL, NULL, NULL, NULL, NULL, NULL);
```

Output

Action Output

| # | Time | Action | Message |
|---|---|---|---|
| 5012 | 11:47:02 | SELECT * FROM transactions.data_user | 5000 row(s) returned |
| 5013 | 11:51:38 | INSERT INTO data_user (id, name, surname, phone, personal_email, birth_date, country, city, postal_code, address) VALUES ('9999', NULL, NULL, N... | 1 row(s) affected |

Un cop existent el registre a les taula *data_user*, ja podem afegir la FK:

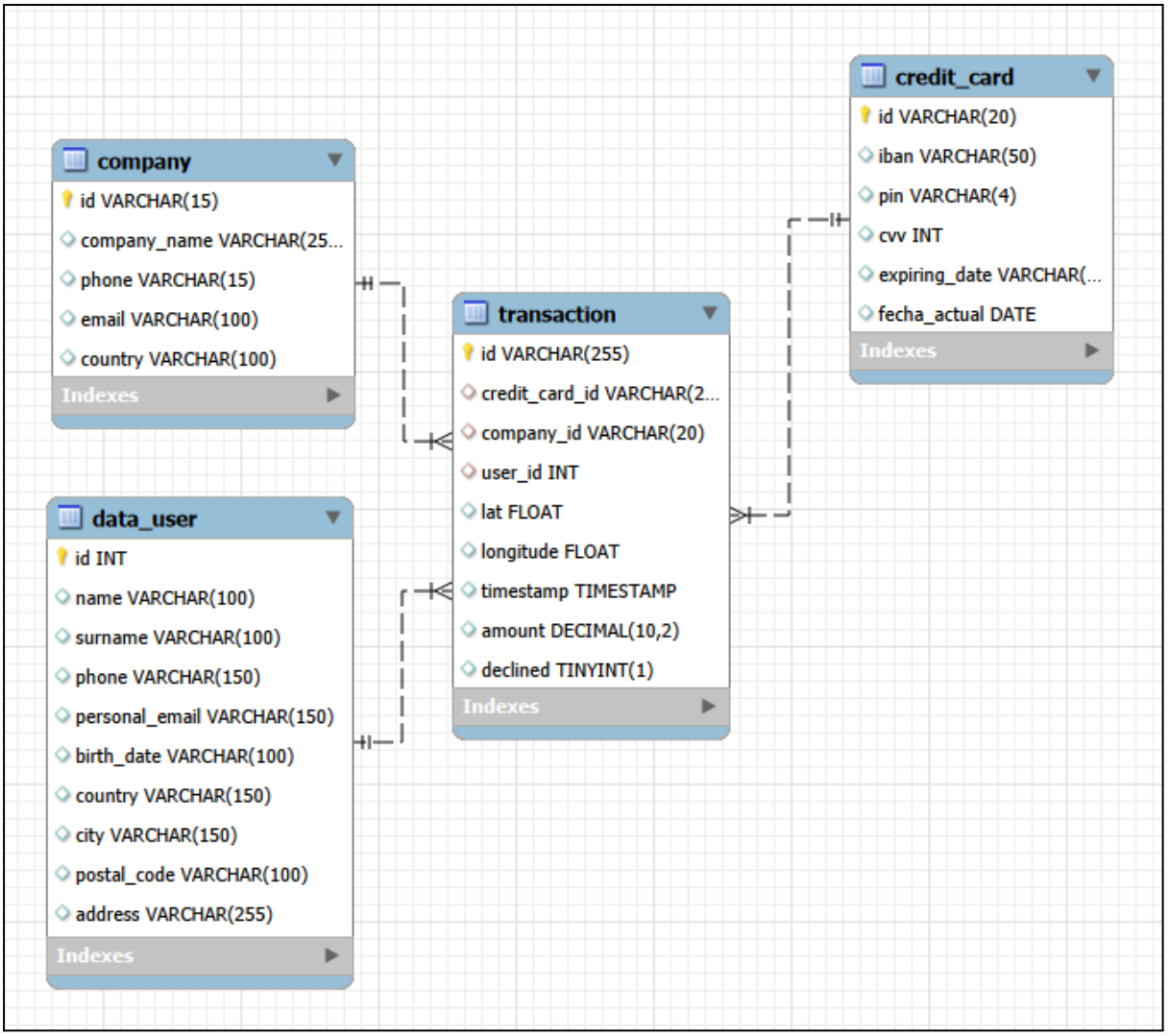
```
#Repeteixo ADD FK
ALTER TABLE transaction
ADD FOREIGN KEY (user_id) REFERENCES data_user(id);
```

Output

Action Output

| # | Time | Action | Message |
|---|---|---|---|
| 5013 | 11:51:38 | INSERT INTO data_user (id, name, surname, phone, personal_email, birth_date, country, city, postal_code, address) VALUES ('9999', NULL, NULL, N... | 1 row(s) affected |
| 5014 | 11:54:05 | ALTER TABLE transaction ADD FOREIGN KEY (user_id) REFERENCES data_user(id) | 100000 row(s) affected Records: 100000 Duplicates: 0 Warnings: 0 |

Ara ja tenim el diagrama tal com ens el demanàvem:




## Exercici 2

L'empresa també us demana crear una vista anomenada "InformeTecnico" que contingui la següent informació:

- ID de la transacció
- Nom de l'usuari/ària
- Cognom de l'usuari/ària
- IBAN de la targeta de crèdit usada.
- Nom de la companyia de la transacció realitzada.
- Assegureu-vos d'incloure informació rellevant de les taules que coneixereu i utilitzeu àlies per canviar de nom columnes segons calgui.

Mostra els resultats de la vista, ordena els resultats de forma descendent en funció de la variable ID de transacció.

Creació de la vista *InformeTecnico*:

```
CREATE VIEW InformeTecnico AS
SELECT
    t.id AS transaction_id,
    du.name AS user_name,
    du.surname AS user_surname,
    cc.iban AS IBAN,
    c.company_name
FROM  transaction t
JOIN data_user du ON t.user_id = du.id
JOIN credit_card cc ON t.credit_card_id = cc.id
JOIN company c ON t.company_id = c.id;
```

Output

Action Output

| # | Time | Action | Message |
|---|---|---|---|
| 5016 | 12:07:05 | CREATE VIEW InformeTecnico AS SELECT t.id AS transaction_id, u.name AS user_name, u.surname AS user_surname, cc.iban AS IBAN, c... | Error Code: 1054. |
| 5017 | 12:07:30 | CREATE VIEW InformeTecnico AS SELECT t.id AS transaction_id, du.name AS user_name, du.surname AS user_surname, cc.iban AS IBAN,... | 0 row(s) affected |

I mostra dels resultats:

```
SELECT *
FROM InformeTecnico
ORDER BY transaction_id DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

| transaction_id | user_name | user_surname | IBAN | company_name |
|---|---|---|---|---|
| FFFD31D6-9495-47CE-B54A-7DB8E1CC2... | Bmrqli | Tprvmrc | XX7948144512112891824909 22 | Turpis Company |
| FFFCF76D-ECF0-4985-A2D0-B2A7B7599... | Dfrled | Vilqcjdl | XX6362517016478920366760 34 | Amet Nulla Donec Corporation |
| FFFC9E8D-27C7-4ADE-98F2-7533EF4DF... | Securp | Faofvqfy | XX1626771433042236314375 67 | Nunc Interdum Incorporated |
| FFFB270D-F53A-4D5D-9666-E5307C53C... | Gqzjpa | Uirzjulh | XX3951142670820199525670 52 | Viverra Donec Foundation |
| FFF9E3CE-234E-408C-A8EF-F9CAD5772... | Yshimq | Zpsjsleed | XX8845462156537570367941 | Convallis In Incorporated |

InformeTecnico 4

Output

Action Output

| # | Time | Action | Message |
|---|---|---|---|
| 5019 | 12:10:30 | SELECT * FROM InformeTecnico ORDER BY transaction_id DESC | 100000 row(s) returned |